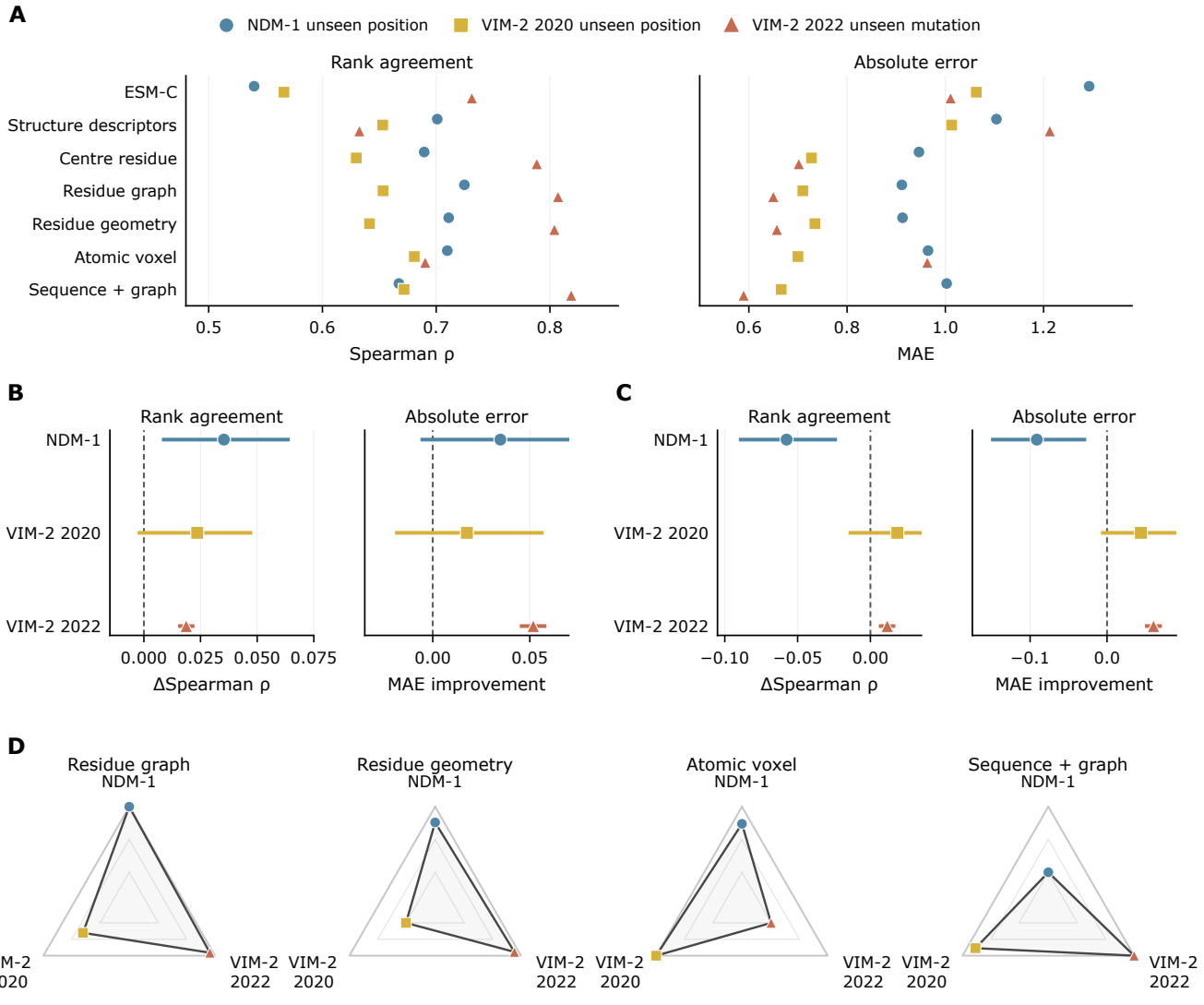

A
NDM-1 unseen position
VIM-2 2020 unseen position
VIM-2 2022 unseen mutation
Rank agreement
Absolute error
ESM-C
Structure descriptors
Centre residue
Residue graph
Residue geometry
Atomic voxel
Sequence + graph
0.5
0.6
0.7
0.8
Spearman ρ
0.6
0.8
1.0
1.2
MAE
B
Rank agreement
Absolute error
NDM-1
VIM-2 2020
VIM-2 2022
0.000
0.025
0.050
0.075
ΔSpearman ρ
0.00
0.05
MAE improvement
C
Rank agreement
Absolute error
NDM-1
VIM-2 2020
VIM-2 2022
−0.10
−0.05
0.00
ΔSpearman ρ
−0.1
0.0
MAE improvement
D
Residue graph
Residue geometry
Atomic voxel
Sequence + graph
NDM-1
VIM-2 2020
VIM-2 2022